Impredicativity: Overview

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What is impredicativity, informally?

"Defining an object while already referring to it"

Some examples:

- Infima and suprema of sets $X \subseteq \mathbb{R}$ of real numbers

$$\inf X = s \qquad \text{iff} \qquad (\forall x \in X.\, x \leq s) \wedge (\forall s' \in \mathbb{R}.\, (\forall x \in X.\, x \leq s') \rightarrow s \leq s')$$

- The problematic set from the Russell paradox

$$R \;:=\; \{X \mid X \notin X\} \qquad \text{with} \qquad R \in R \leftrightarrow R \notin R$$

What is impredicativity, formally?

In our type-theoretic setting, impredicativity can come in various strengths:

1. We could impose Type : Type
   - Strongest form of impredicativity, simply treating Type as a normal type
   - Immediately inconsistent, i.e. induces an inhabitant of the $\bot$ type

2. We could have a impredicative typing rules for Type
   - For instance products $\Pi x : T.U$ have kind Type even if T is Type itself
   - Consistent when restricting inductive types and classical axioms

3. We could have an additional impredicative universe Prop
   - Products $\Pi x : T.U$ have type Prop whenever U has type Prop
   - Consistent with classical axioms when severely restricting inductive types

The radical version: Type is a Type

Formally, this amounts to adding a typing rule as follows:

$$\frac{\Gamma \text{ wf}}{\Gamma \vdash \text{Type} : \text{Type}}$$

We can derive the impredicative product rule as an instance of the normal product rule:

$$\frac{\Gamma \vdash \text{Type} : \text{Type} \qquad \Gamma(T : \text{Type}) \vdash U : \text{Type}}{\Gamma \vdash \Pi T : \text{Type}.\, U \;:\; \text{Type}}$$

However, we can now also define some very problematic types:

$$\frac{\vdash \text{Type} : \text{Type} \qquad \dfrac{(T : \text{Type}) \vdash T : \text{Type} \qquad (T : \text{Type})(x : T) \vdash \text{Type} : \text{Type}}{(T : \text{Type}) \vdash T \to \text{Type} : \text{Type}}}{\vdash \Sigma T : \text{Type}.\, T \to \text{Type} \;:\; \text{Type}}$$

We will see later how this type can be exploited for a paradox...

The moderate version: Type is impredicative

A much more restricted approach is to just have the impredicative typing rules:

$$\frac{\Gamma \vdash T : s \qquad \Gamma(T : s) \vdash U : \mathrm{Type}}{\Gamma \vdash \Pi x : T.\, U \;:\; \mathrm{Type}} \qquad \frac{\Gamma \vdash T : s \qquad \Gamma(T : s) \vdash U : \mathrm{Type}}{\Gamma \vdash \Sigma x : T.\, U \;:\; \mathrm{Type}}$$

In this setting, we can additionally have:

- Large inductive types like ENS : Type with sup : $\Pi\, T : \mathrm{Type}.\, (T \to \mathrm{ENS}) \to \mathrm{ENS}$
- Small inductive types with large eliminations, e.g. $0 \neq 1$ still derivable for $N : \mathrm{Type}$

However, we cannot have:

- Large eliminations for large inductive types, e.g. $\pi_1(\sup\, T\, f) := T$ is not definable
- Classical logic in the form $\Pi\, T : \mathrm{Type}.\, T + (T \to \bot)$, as we will see next lecture

The careful version: an impredicative Prop universe

Introduce additional sort Prop : Type with impredicative typing rules:

$$\frac{\Gamma \vdash T : s \qquad \Gamma(T : s) \vdash U : \mathrm{Prop}}{\Gamma \vdash \Pi x : T.\, U \;:\; \mathrm{Prop}} \qquad \frac{\Gamma \vdash T : s \qquad \Gamma(T : s) \vdash U : \mathrm{Prop}}{\Gamma \vdash \Sigma x : T.\, U \;:\; \mathrm{Prop}}$$

Interpreting Prop as proof-irrelevant, we can additionally have:

- Large inductive types like ENS : Prop with sup : $\Pi\, T$: Prop. $(T \to \mathrm{ENS}) \to \mathrm{ENS}$
- Classical logic in the form $\Pi\, T$: Prop. $T + (T \to \bot)$, or even stronger assumptions

However, we cannot have:

- Large eliminations for most inductive types, e.g. $0 \neq 1$ not derivable for N : Prop